3. Aero Balance

→ How total downforce ~~sp~~ is split b/w front and rear axle

→ Why it Matters?

↳ More downforce on front allows

- More grip for front tyres
- Front bites harder
- Car rotates aggressively. (High speed oversteer)

↳ More downforce on rear:

- Rear grip is high
- High speed understeer.

→ How this affects LTD? (Lap Time Decomposition)

↳ If front aero is too strong:

- Earlier lift in fast corners.
- Rear instability limits apex speed.

↳ If rear aero ~~is~~ is too strong.

- Car won't rotate
- Mid corner speed lower.
- More steering angle - more tyre wear.

Why aero matters in Mid Corner ?

→ What happens ?

↳ In mid corner

- Constant relative speed
- Stedy steering angle
- Peak lateral acceleration

↳ At high speed Down force increases – increasing lateral grip

- higher sustainable corner speed.

Aero load equals or exceeds car's weight, cornering capability is dominated by aerodynamics.

5 Drag and Straight.

→ Core trade off.

↳ Less wing :

- Less downforce
- Less drag
- Faster on straight.

↳ At straight drag reduces acceleration, top speed hence costing speed.

→ Straight line low.

↳ Loosing speed at straight depends on 2

1. Exit Limited
2. wing Limited.

↳ Exit Limited Low.

- No drag.
- Happens because exit from corner became slower.

↳ Wing Limited Low.

- Drag induced via wings, cooling, floor sensitivity.

6 Dirty Air

↳ When a car drives through air it leaves behind:

- Turbulance
- ~~Swi~~ Swirling vortex
- Lower energy air flow.

This messes with aero, of the following car.

→ What Actually Happens?

↳ Front wing is designed for

- Smooth
- High energy
- Predictable flow.

↳ When dirty ~~air~~ air hits.

- Front downforce drops.
- Aero balance ~~sh~~ shifts.
- Car develops high speed understeer.

↳ This affects mid-corner of the following driver

- Has to lift earlier
- Less maximum speed
- Lower mid corner speed

This affects the exit, which compounds in next straight.